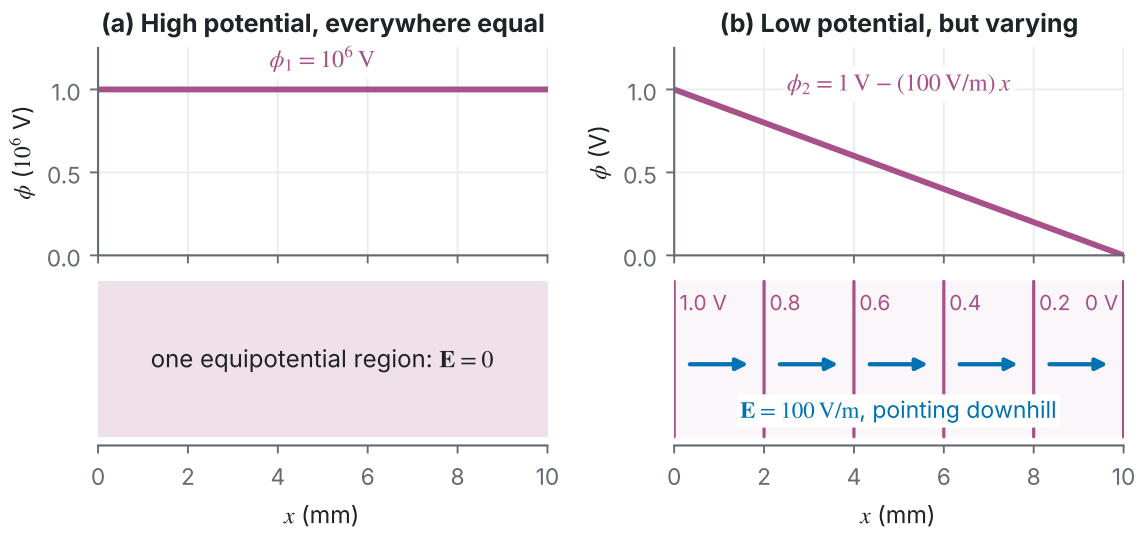
(a) High potential, everywhere equal
$\phi_1 = 10^6$ V
1.0
0.5
0.0
ϕ (10^6 V)
one equipotential region: $\mathbf{E} = 0$
0
2
4
6
8
10
x (mm)
(b) Low potential, but varying
$\phi_2 = 1\text{ V} - (100\text{ V/m})\,x$
1.0
0.5
0.0
ϕ (V)
1.0 V
0.8
0.6
0.4
0.2
0 V
$\mathbf{E} = 100$ V/m, pointing downhill
0
2
4
6
8
10
x (mm)